

SALES TREND AND TIME-BASED PERFORMANCE ANALYSIS FOR AFFICIONADO COFFEE ROASTERS

Research Paper

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Abstract

This research paper presents a comprehensive sales trend and time-based performance analysis for Afficionado Coffee Roasters using transaction-level retail data. The project focuses on understanding customer purchase behavior across various hours, days, and store locations. The study identifies peak transaction hours, busiest days, hourly demand patterns, and store-level performance variations. The analysis was performed using Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Plotly, and Streamlit. The project aims to help management optimize staffing schedules, improve operational planning, and enhance customer experience through data-driven decision-making.

1. Introduction

Retail businesses generate a significant amount of transactional data daily. However, organizations often fail to convert raw data into meaningful insights that can improve operational efficiency. In specialty coffee retail businesses such as Afficionado Coffee Roasters, understanding customer demand patterns is essential for staffing optimization, inventory management, and customer satisfaction.

This project focuses on analyzing transaction-level coffee sales data to identify important temporal patterns such as busiest transaction hours, peak sales periods, and store-wise customer demand behavior. The findings of this analysis help the organization transition from intuition-based decisions to evidence-based operational planning.

2. Problem Statement

Afficionado Coffee Roasters currently lacks a consolidated analytical system for understanding sales patterns across different periods of the day and week. Management faces several operational challenges such as:

- Overstaffing during slow business hours
- Understaffing during peak demand periods
- Inefficient workforce allocation
- Limited understanding of store-level demand patterns
- Reactive operational decision-making

Without proper analytics, the organization cannot effectively optimize customer service quality or operational costs.

3. Objectives

Primary Objectives:

- Identify overall sales trends.
- Determine busiest and slowest days of the week.

- Identify peak transaction hours.

Secondary Objectives:

- Compare temporal demand patterns across store locations.
- Support workforce scheduling and operational planning.
- Generate business recommendations using analytical insights.
- Create interactive dashboards for decision support.

4. Scope of the Project

The scope of this project includes sales trend analysis, hourly demand analysis, day-wise revenue analysis, store-level comparison, and time bucket analysis using transaction-level coffee retail data. The project focuses on deriving actionable business insights that improve operational efficiency and customer experience.

5. Dataset Description

The dataset used in this project contains transactional information related to coffee sales operations.

Total Records: 149116

Total Columns: 11

Dataset Attributes:

- transaction_id – Unique transaction identifier
- transaction_time – Timestamp of transaction
- transaction_qty – Quantity purchased
- unit_price – Product price per unit
- store_id – Store identifier
- store_location – Physical store location
- product_id – Product identifier
- product_category – Product category
- product_type – Product variant
- product_detail – Detailed product description

6. Tools and Technologies Used

The following technologies and tools were used during the project implementation:

- Python – Core programming language used for analysis
- Jupyter Notebook – Interactive development environment
- Pandas – Data cleaning and manipulation
- NumPy – Numerical computations

- Matplotlib – Visualization library
- Seaborn – Statistical data visualization
- Plotly – Interactive chart creation
- Streamlit – Dashboard development
- Excel – Source data storage format

7. Data Preprocessing and Validation

Before performing analysis, several preprocessing and validation steps were applied to ensure data quality and consistency.

The following preprocessing activities were performed:

- Loading the Excel dataset into Pandas DataFrame
- Checking for missing values
- Identifying duplicate records
- Validating transaction quantities and prices
- Converting transaction_time into datetime format
- Removing inconsistent records if necessary

These preprocessing steps helped improve analytical accuracy and ensured reliable business insights.

8. Feature Engineering

Feature engineering was performed to derive additional meaningful attributes from the raw dataset.

Derived Features:

- Revenue = transaction_qty × unit_price
- Hour of transaction
- Day name
- Time buckets:
 - Morning (6–11)
 - Afternoon (12–16)
 - Evening (17–21)
 - Late Hours (22–5)

These features enabled detailed temporal analysis and customer behavior interpretation.

9. Methodology

The following analytical methodology was followed during the project:

Step 1: Data Collection

The transaction-level dataset was collected from the company database in Excel format.

Step 2: Data Cleaning

Missing values, duplicates, and invalid records were identified and handled appropriately.

Step 3: Feature Engineering

Additional features such as revenue, hour, day name, and time bucket were generated.

Step 4: Exploratory Data Analysis

Visualization techniques were applied to identify sales patterns and demand trends.

Step 5: Business Insight Generation

Analytical findings were interpreted from a business operations perspective.

Step 6: Dashboard Development

Interactive visual dashboards were created using Streamlit for operational monitoring.

10. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer transaction behavior and sales patterns.

The following analyses were conducted:

- Revenue distribution analysis
- Hourly transaction volume analysis
- Revenue by day of week
- Store-level sales comparison
- Time bucket demand analysis
- Heatmap visualization for temporal demand patterns

Visualization techniques such as line charts, bar charts, and heatmaps were used to identify meaningful business patterns.

11. Sample Python Implementation

```
import pandas as pd
```

```
df = pd.read_excel("Afficionado Coffee Roasters.xlsx")
```

```
df['Revenue'] = df['transaction_qty'] * df['unit_price']
```

```
df['transaction_time'] = pd.to_datetime(df['transaction_time'])
```

```
df['Hour'] = df['transaction_time'].dt.hour

df['Day_Name'] = df['transaction_time'].dt.day_name()

def time_bucket(hour):
    if 6 <= hour <= 11:
        return 'Morning'
    elif 12 <= hour <= 16:
        return 'Afternoon'
    elif 17 <= hour <= 21:
        return 'Evening'
    else:
        return 'Late Hours'

df['Time_Bucket'] = df['Hour'].apply(time_bucket)
```

12. Key Findings and Insights

The analysis revealed several important operational insights:

- Morning hours generated the highest customer traffic and revenue.
- Weekdays showed higher transaction activity compared to weekends.
- Some store locations experienced strong evening demand patterns.
- Late-night transaction volumes were significantly lower.
- Revenue concentration during specific hours suggests the need for dynamic staffing schedules.
- Certain product categories contributed significantly to overall sales performance.

13. Recommendations

Based on the analytical findings, the following recommendations are proposed:

- Increase staffing during peak morning hours.
- Reduce workforce allocation during low-demand periods.
- Launch promotional offers during slow business hours.
- Adjust inventory distribution according to store demand patterns.
- Use real-time dashboards for continuous sales monitoring.
- Implement data-driven operational planning strategies.

14. Streamlit Dashboard

An interactive Streamlit dashboard was developed as part of this project to provide live analytical insights.

Dashboard Features:

- Overall sales trend dashboard
- Day-wise performance charts
- Hourly demand analysis
- Store-level comparison
- Interactive filters for location and time range
- Revenue and quantity analysis toggle

The dashboard improves accessibility of insights for management and operational teams.

15. Benefits of the Project

This project provides several operational and strategic benefits to the organization:

- Improved workforce scheduling
- Better operational efficiency
- Reduced unnecessary labor costs
- Enhanced customer service experience
- Faster business decision-making
- Better understanding of customer behavior patterns

16. Future Enhancements

The project can be extended further with advanced analytics capabilities such as:

- Machine learning-based sales forecasting
- Predictive staffing optimization
- Customer segmentation analysis
- Product recommendation systems
- Real-time streaming analytics
- AI-based operational automation

17. Conclusion

This project successfully demonstrated how data analytics techniques can be used to analyze temporal sales patterns for Afficionado Coffee Roasters. The analysis provided meaningful insights regarding customer demand behavior, peak sales periods, and operational inefficiencies. By implementing the recommendations provided in this study, the organization can improve staffing efficiency, reduce operational costs, and enhance

customer satisfaction. The project also highlights the importance of data-driven decision-making in modern retail business operations.

18. References

- Python Documentation – <https://docs.python.org/>
- Pandas Documentation – <https://pandas.pydata.org/>
- Matplotlib Documentation – <https://matplotlib.org/>
- Seaborn Documentation – <https://seaborn.pydata.org/>
- Streamlit Documentation – <https://streamlit.io/>